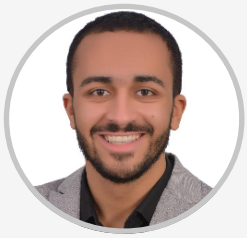
europass



Mostafa Ahmed

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ABOUT ME

Robotics engineer and Erasmus Mundus M.Sc. student in Intelligent Field Robotic Systems, with experience in reinforcement learning and MPC (BSK-RL and acados trajectory optimization in Basilisk), ROS 2 mobile robots and manipulators, and physics-based simulation (Gazebo, MuJoCo, Isaac Sim). Worked with real and simulated platforms including TurtleBot systems, industrial arms, and space-robotics SIL. Research interests include robot learning and optimal control for locomotion and interaction, sim-to-real transfer, and controllers that remain reliable on hardware.

EDUCATION AND TRAINING

01/09/2025 – CURRENT Spain

M.SC. IN INTELLIGENT FIELD ROBOTIC SYSTEMS (ERASMUS MUNDUS JOINT MASTER) Joint Degree: Univ. of Girona (UdG), Eötvös Loránd Univ. (ELTE), Univ. of Zagreb (UniZG)

Autonomous Systems, Probabilistic Robotics, Machine Learning, 3D Vision, and Research Methodologies.

15/09/2020 – 15/03/2025 Alexandria, Egypt

B.SC. IN MECHATRONICS AND ROBOTICS ENGINEERING Egypt-Japan University of Science and Technology (E-JUST)

Robotics, Control Systems, AI, Embedded Systems, Mechanical Design, and Software Engineering.

Final grade CGPA 3.35 / 4.00 (Very Good)

WORK EXPERIENCE

SPACE ROBOTICS AND GNC INTERN – ADVANCED CENTER FOR AEROSPACE TECHNOLOGIES (CATEC) – 01/06/2026 – 31/08/2026 – SEVILLA, SPAIN

- Developed and evaluated spacecraft robotics simulation workflows for Guidance, Navigation, and Control (GNC), docking, rendezvous, contact dynamics, optical navigation, and autonomy.
- Built Software-in-the-Loop (SIL) benchmark scenarios using Gazebo, MuJoCo, NVIDIA Isaac Sim, Basilisk, and Space ROS for robotic manipulation and spacecraft control validation.
- Executed mission design and cross-tool physics benchmarking across GMAT, FreeFlyer, NASA 42, Basilisk, and Orekit, producing metrics, plots, and technical reports for simulation tool selection.

RESEARCH ASSISTANT – CONTROL LAB, EGYPT-JAPAN UNIVERSITY OF SCIENCE AND TECHNOLOGY (E-JUST) – 01/09/2024 – 28/02/2025 – ALEXANDRIA, EGYPT

- **FPGA Integration:** Developed a LabVIEW I2C subroutine for NI sbRIO FPGA to evaluate mechatronic sensor performance.
- **Academic Research:** Co-authored a comparative study on sensor costs, currently under peer review.

TEAM LEADER – E-JUST RACING TEAM (STUDENT ENGINEERING TEAM) – 01/04/2022 – 01/02/2025 – EGYPT

- **Leadership & EV Systems**: Led cross-functional teams in EVER and Shell Eco-marathon, mentored junior engineers, and directed electric vehicle design/manufacturing, achieving 6th place in EVER.

ELECTRICAL AUTOMATION ENGINEER INTERN – NESTLÉ – 15/07/2023 – 15/09/2023 – GIZA, EGYPT

- **Automation**: Programmed Allen-Bradley PLCs, integrated OPC tags with DMO platform, and supported hardware retrofits for 40+ obsolete components, reducing downtime by 20%.

MECHATRONICS DESIGN ENGINEER INTERN – BMC ALMASRYA FOR ADVANCED INDUSTRIES – 01/11/2023 – 01/12/2023 – ALEXANDRIA, EGYPT

- **Machine Design**: Designed an automated forming machine in SolidWorks with pneumatic gating and deployed a PLC-based control system to improve production reliability.

MECHATRONICS MAINTENANCE ENGINEER INTERN – ELWELELY FOR AGRICULTURAL CROPS – 01/05/2023 – 30/06/2023 – ALEXANDRIA, EGYPT

- **Maintenance & PLCs**: Troubleshot 10+ industrial systems and upgraded 3 relay-based setups to Schneider PLC/HMI control, reducing breakdowns by 25%.

SKILLS

Robotics

ROS 2 | ROS | RViz | Gazebo | SLAM | EKF | sensor fusion | mobile manipulation | motion planning

Manipulation & Control

Task-priority control | WDLS | Inverse Kinematics | PID | LQR | Trajectory planning | Resolved-rate control

Simulation

MATLAB/Simulink | LabVIEW | SolidWorks | ANSYS | MuJoCo | Basilisk | Space ROS | NVIDIA Isaac Sim

Perception

OpenCV | YOLO | ArUco | COLMAP | Structure-from-Motion | Multi-View Stereo

Programming

Python | C/C++ | MATLAB | Linux | Version Control (Git) | NumPy | PLC programming | STM32 | CAN | I2C

PROJECTS

01/02/2026 – 30/06/2026
Vehicle-Manipulator System

- Built a ROS 2 mobile manipulation system for a TurtleBot-based robot with a SwiftPro arm, integrating kinematic modeling, forward kinematics validation, resolved-rate motion control, and weighted damped least-squares inverse kinematics.
- Implemented recursive task-priority control for constrained manipulation, including end-effector tracking, joint-limit handling, base-arm coordination, and configurable task hierarchies.
- Developed an FSM-based pick-and-place/intervention pipeline using ArUco marker perception, TF2 frame transforms, gripper control, and ROS 2 visualization/debugging tools.
- **Tools**: ROS 2, Python, NumPy, TF2, ArUco, inverse kinematics, task-priority control, mobile manipulation.

Industrial Robotic Manipulation Systems (UR3e, TX60, TS60)

- **Industrial Automation (Stäubli):** Programmed TX60 (6-DOF) and TS60 (SCARA) robots using VAL3 and Stäubli Robotics Suite for high-speed assembly, de-palletizing, and precise trajectory execution.
- **Collaborative Robotics (Universal Robots):** Developed safe, sensor-integrated motion planning and force-controlled polishing applications on a UR3e utilizing PolyScope.

03/2023 – 10/2023
Minesweepers – Autonomous Vehicle-Manipulator System

- **Global Award**: Secured 2nd place globally by developing a ROS-based vehicle-manipulator robot for surface landmine detection, localization, and collection.
- **Manipulation**: Integrated a robotic arm with an electromagnetic end-effector for safe mine pickup and multi-point collection.
- **Perception & Control**: Implemented YOLO detection, EKF localization, IMU fusion, teleoperation, RViz visualization, and a custom mapping GUI.
- **Tools**: ROS, Python, Raspberry Pi, OpenCV, YOLO, RViz, EKF, IMU.

01/02/2026 – 30/06/2026
Graph-SLAM Localization & Kinodynamic Exploration for TurtleBot 4

- Implemented a ROS 2 Jazzy autonomy stack combining covariance-triggered Graph SLAM, GTSAM iSAM2 pose-graph optimization, NDT scan matching, loop-closure detection, and RViz-based uncertainty visualization.
- Integrated optimized Graph-SLAM localization with frontier exploration, kinodynamic RRT planning, occupancy mapping, and pure-pursuit path following for TurtleBot 4 navigation.
- **Tools**: ROS 2 Jazzy, C++, Python, GTSAM, PCL, Eigen, RViz, slam_toolbox, TurtleBot 4.

01/02/2026 – 30/06/2026

Underwater Structure-from-Motion & Factor-Graph Reconstruction

- Developed an underwater perception pipeline for Girona 500 / Stonefish AUV data, combining COLMAP visual reconstruction, loop-closure-aware keyframe selection, and GTSAM factor-graph optimization over 3,641 odometry frames and 279 visual keyframes.
- Fused noisy DVL/INS odometry with robust visual reprojection constraints, reducing mean reprojection error from 22.1 px to 0.47 px and generating a dense Poisson mesh with 3.67M fused points and 10.6M vertices.
- **Tools**: Python, ROS bags, COLMAP, GTSAM, Open3D, NumPy, Structure-from-Motion, Multi-View Stereo.

HONOURS AND AWARDS

03/2023

1st Place Global Winner, Lunar Lander Extended Reality Challenge (ASME)

Link https://drive.google.com/file/d/101gxYfzW9zlZOfWYiZump0Pv5-DdXAnq/view?usp=sharing

PUBLICATIONS

2025

Enhancing AUTOSAR-Based Firmware Over-the-Air Updates in the Automotive Industry with a Practical Implementation on a Steering System

Formal publication detailing the **Valeo Best Project Award-winning** secure FOTA system.

Bachelor's Graduation Thesis

Authors: Mostafa A. Mostafa, et al. | **Publisher**: Egypt-Japan University of Science and Technology

2024

Automating Hot-Rolling: Designing an Integrated Mechatronics System for Enhanced Efficiency in Sheet Metal Production

Modeled multi-body dynamics (AC motor, power screw) in **MATLAB/Simulink** and designed a **PID controller** to automate sheet metal hot-rolling speed and thickness.

Authors: Mostafa A. Mostafa, et al.

2023

Implementing PID Controller on a Hand Follower Robot Using Numerical Methods

Developed a **PID controller** using **numerical methods** and **auto-tuning** techniques for a hand-follower robot to optimize system stability and eliminate steady-state error.

Authors: Mostafa A. Mostafa, et al.